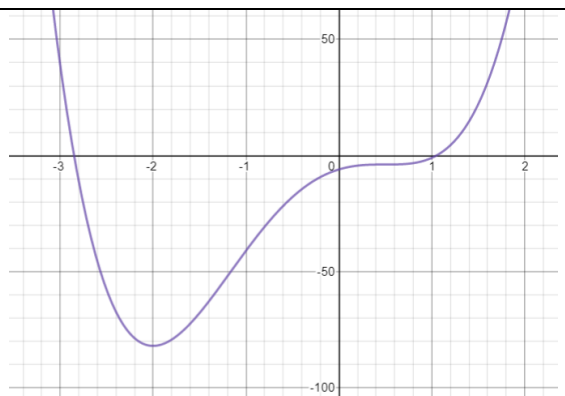
Question			Answer	Marks	AO	Guidance
16	**(a)**		$\frac{dy}{dx} = 24x^3 + 24x^2 - 42x + 12$	**M1**	**3.1a**	allow one sign or coefficient error; must be four terms
			their $\frac{dy}{dx} = 0$	**M1**	**1.1**	at least two terms correct
			f(k) evaluated, where k is a factor of ±12 or $\pm\frac{a}{12}$, where a = 1,2,3,4 or 6	**M1**	**2.1**	may be implied by $x = -2$ seen unsupported or $(x + 2)$ identified as factor
			$(x + 2)(4x^2 - 4x + 1)$ or $(2x - 1)(2x^2 + 3x - 2)$	**M1**	**3.1a**	by inspection or long division; allow one sign error or one coefficient error in trinomial may be implied by $x = ½$ seen unsupported or $(2x - 1)$ **oe** identified as factor
			$x = -2$ and $x = \frac{1}{2}$ and no others	**A1**	**1.1**	may see $x = \frac{1}{2}$ (repeated) **A0** for $x = -2$ (repeated)
			$\left(\frac{1}{2}, -\frac{31}{8}\right)$ and $(-2, -82)$ and no others	**A1**	**1.1**	
			$\frac{d^2y}{dx^2} = 72x^2 + 48x - 42$	**M1***	**1.1**	allow one sign or one coefficient error, **FT** their $\frac{dy}{dx}$; allow **M1** for $12x^2 + 8x - 7$
			$\frac{d^2y}{dx^2} = 150$ when $x = -2$ so minimum value	**A1**	**1.1**	**NB** test indecisive at $x = ½$ **A0** for just eg $\frac{d^2y}{dx^2} > 0$ so minimum
			or eg x : –2.1 | (–2) | –1.9 $\frac{dy}{dx}$: –16.224 | (0) | 13.824			award **M1A1** for consideration of gradient either side of –2, values must be correct to at least 2sf for **A1**

Question			Answer	Marks	AO	Guidance
			eg x | 0 | (½) | 1 $\frac{dy}{dx}$ | 12 | (0) | 18 dependent on at least two terms correct in derivative; must see values	**M1**	**3.1a**	or eg x | 0 | (½) | 1 y | –6 | (–3.875) | –1 or eg x | 0 | (½) | 1 $\frac{d^2y}{dx^2}$ | –42 | (0) | 78
			inflection at $\left(\frac{1}{2}, -\frac{31}{8}\right)$	**A1**	**3.2a**	values in table must be correct
			their $72x^2 + 48x - 42 = 0$	**M1dep***	**1.1**	
			$x = -\frac{7}{6}$ **isw**	**A1**	**1.1**	ignore calculation of associated y-value allow any correct decimals to 3 sf or more
				[12]		
16	**(b)**			**M1**	**1.1**	curve with a minimum in 3rd quadrant and stationary point of inflection in 4th quadrant and no other stationary points
				B1	**1.1**	(0, –6) identified as y-intercept (intercept must be below the x-axis and above –20)
				A1	**1.1**	correct curve with intercepts at $(-a,0)$ and $(b,0)$, where $-3 < a < -2.6$ and $0.8 < b < 1.2$; minimum at $(-2, y)$ where $-90 < y < -80$ and inflection for $0 < x < 1$ and y is between the x-axis and the y-intercept
				[3]		